

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 34

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Ship rudders that turn in tighter circles / aerogenerator blades that start turning at lower wind speeds • Water-repelling surface of lotus flowers • Coatings for aircraft and ships that reduce fuel consumption / hospital coatings that bacteria cannot stick to • Hooked burrs on seeds that stick to animal fur	4
01.2	• The mound absorbs sunlight and warms; hot air inside rises, creating a convection current that draws stale air out of the lower parts (solar-driven ventilation) • Air blowing over the mound has a lower pressure which also draws air out • Copied in shopping complexes / office blocks to give natural ventilation and cooling without air conditioning, so less energy / electricity is used	3
01.3	<i>Any 2 from:</i> • Marsupials: young develop in the pouch so are easier to study than a human foetus, helping understanding of developmental problems • Hippopotamus skin secretes hipposudoric acid (natural sunscreen and antimicrobial) which may improve treatment of burns • Marine sponges produce proteins that prevent rejection of grafts, being developed to prevent rejection of transplanted organs	2
01.4	• They are light but strong: hollow with internal struts that prevent bending / breaking, copied in lightweight bridges and roofs	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11-14; AQA spec 3.1.2.1.1.	
02.1	• Spearman's rank correlation coefficient • Chi-squared test (frequencies / counts in categories) • t-test (two means of measured data) • Spearman's rank correlation coefficient	4

Question	Marking guidance	Marks
02.2	Chi-squared compares frequencies / counts in categories; it cannot be used for continuously variable measured data	1
02.3	- The value (ignoring sign) exceeds the critical value so the correlation is significant at the 5% level - The negative sign shows a negative correlation: one variable decreases as the other increases - There is less than a 5% probability the correlation arose by chance; reject the null hypothesis (but correlation is not proof of causation)	3
02.4	- Show the standard deviation / range of the original values around each mean - If the bars of two means overlap greatly the difference is probably not significant; little overlap suggests a significant difference	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	
03.1	- Species: a group of closely related organisms that naturally interbreed to produce fertile offspring and are reproductively isolated from other groups (<i>Passer domesticus</i>, house sparrow); the only 'real' taxon - Niche: the role a species plays in its habitat - how it uses resources and relates to other species (tawny owl: nocturnal carnivore nesting in tree holes eating small mammals); two species cannot share exactly the same niche in one habitat - Community: all the populations of all species (plants, animals, fungi, bacteria, protists, archaea) living in an area - Biome: a large geographical region with a specific climate and characteristic community, including all areas where it occurs (all tropical rainforests, tundra, savannah)	8
03.2	- An ecosystem is the biotic community plus abiotic factors of one area, relatively self-contained (a particular coral reef) - A biome includes all separate ecosystems with the same climate and community type (all coral reefs); each reef is a different ecosystem within the same biome	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 82-85; AQA spec 3.1.3.2.	
04.1	- Very small gene pool so individuals are closely related and inbreeding is likely - Harmful recessive genes are expressed when both parents carry them: some goslings had thin hair-like feathers with poor insulation so survival in cold weather was low - A stud book records family trees so breeding is arranged between the least related individuals (often coordinated by a specialist zoo) - Exchange of individuals / semen between zoos; avoid hybridisation with related species / subspecies	4
04.2	<i>Any 3 from:</i> - Breeding is triggered by stimuli (day length, light, temperature, food, body fat) which may be unknown / not provided - Population interactions: mixing breeding and non-breeding individuals causes conflict; some need a choice of partners, others pair for life - Breeding habitat needed: courtship display sites, nest sites, social groups (flamingos only breed in large groups - mirrors used) - Stress / disturbance	3

Question	Marking guidance	Marks
04.3	• AI: semen collected (and can be stored / cryopreserved for years) and inserted into the female, so animals need not be transported or meet; avoids dangerous mating (elephants, rhinos) and partner rejection • Embryo transfer: a female of the endangered species (eg bongo) is treated with hormones to release many eggs, which are fertilised and implanted into surrogate females of a common related species (eland) • So many offspring are produced per cycle instead of one calf per year, overcoming the shortage of breeding females	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 33-35; AQA spec 3.1.2.3.3.	
05.1	• In winter nutrients are high (storms disturb sediments, deep mixing) but light is too low for photosynthesis • In spring light rises while nutrients are still high so growth is rapid - the spring bloom (peak about May) • The bloom uses up nutrients (falling to 10 by July); dead plankton sink out of the photic zone carrying nutrients to the seabed • So nutrients become limiting in summer despite high light; a smaller autumn bloom follows when storms return nutrients while light is still adequate	4
05.2	• Warm surface water is stable (no seasonal mixing / storms) so nutrients lost to deep water in sinking dead organisms are not replaced • Nutrient (especially phosphate) concentrations in the photic zone remain very low - the limiting factor	2
05.3	• Upwelling (offshore winds, seamounts, currents deflected upwards) brings cold nutrient-rich deep water into the photic zone • Coastal waters receive nutrients washed off the land in rivers and storms disturb shallow sediments • Nutrients plus light in the shallow photic zone allow high algal productivity supporting rich food webs and fisheries (eg Peru anchovy)	3
05.4	• Phosphates have low solubility and no atmospheric reservoir; they are only added by river runoff or upwelling and are lost to sediments	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-339; AQA spec 3.5.2.1.	
06.1	• $1000 / 100\,000 \times 100 = 1\%$	1
06.2	• Some light is reflected by the leaves • Some light is transmitted (passes through the leaves without being absorbed) or is of wavelengths not absorbed • Much of the energy captured is used in the plant's metabolic processes and released as low energy-density heat (respiration)	3
06.3	• Less energy is passed to each trophic level than was received from the previous one (heat losses in respiration) • After four levels too little energy remains to support a further population	2
06.4	• Permanent grassland is a more sustainable land use than arable crops in these areas (soil / rainfall unsuitable for crops) • Grazing animals can digest grass cellulose using symbiotic bacteria in their stomachs, producing food that humans can eat	2

Question	Marking guidance	Marks
06.5	• Pigs are omnivores that can eat food wastes not wanted by people (eg cheese-making wastes, catering leftovers) • So edible meat is produced from material that would otherwise be wasted, without using land / crops that could feed people	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
07.1	• Drainage of wetlands / lowering water tables by groundwater abstraction dries out wetlands; HEP schemes alter river levels flooding sandbank nests of turtles / birds; species with narrow tolerance (specialists) affected first; sundews lost when drained soil becomes aerobic / nitrogen-rich and taller competitors invade • Hot water discharges and organic wastes (sewage) reduce dissolved oxygen, killing aerobic organisms such as fish and insect larvae • Acid mine drainage and acid rain gases lower pH, denaturing proteins of exposed tissues (fish eggs, gills) and dissolving calcium exoskeletons (crayfish)	6
07.2	• Reduces light penetration so submerged plants cannot photosynthesise • Blocks the gills of filter feeders such as bivalve molluscs; sediment smothers the bed	2
07.3	• Rare species often have very specialised habitat requirements / narrow ranges of tolerance • A small change moves conditions outside their range while generalists survive; small populations cannot recover	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 22-23; AQA spec 3.1.2.2.	
08.1	• All greenhouse gases expressed as CO₂ equivalent (allowing for different warming strengths) • Direct emissions from fuel combustion plus indirect: manufacture of components, maintenance items, final disposal (life-cycle analysis) • A fair share of the infrastructure: roads, car parks, signals, street lighting	3
08.2	• Compares the populations of over 18 000 populations of 3700 vertebrate species (terrestrial, freshwater, marine) with 1970 levels • Published in the Living Planet Report as a measure of the state of the planet's ecosystems / biodiversity loss over time	2
08.3	• Hardy, disease-resistant Boran cattle; regular tick dipping reduces ticks for wild herbivores too • Mobile bomas (night enclosures) protect cattle from lions and leave dung that enriches soil and plant growth • Electric fences keep game in and poachers out; biological corridors let animals (except rhinos) move to other reserves; problem elephants relocated • Monitoring: radio-collared lions, drones, camera traps, track surveys; endangered species (Grevy's zebra, northern white rhino) protected • Community links: ecotourism income funds health, education and water projects; local employment; cattle breeding help for local farmers	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 380-383; AQA spec 3.6.4.2.	
09.1	• B	1

Question	Marking guidance	Marks
09.2	<i>Any 4 from:</i> • Harvesting no faster than the MSY - resource never depleted • Mixed species plantations - more food choices, relationships and niches • Indigenous species - support indigenous insects, birds • Mixed age structure - varied light, temperature, wind so more species; continuous cover and yield • Selective logging - small clearings, less disturbance, long-term timber benefit • Retention of dead wood, buffer zones, corridors	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Protected areas: Korup NP Cameroon (1260 km², 600 tree species, endemic <i>Ancistrocladus korupensis</i> with anti-HIV / antimalarial alkaloids); Tumucumaque NP Brazil (39 000 km², adjoining French Guiana reserve); Alto Maues; Rio Bravo Belize funded by charities, companies and ecotourism • Debt for nature swaps; sustainable exploitation: selective logging of high-value timber (mahogany) funding conservation; ITTO aims (largely unmet); CITES listing of timber species (Honduras mahogany Appendix II) • Ecotourism income (but visitor damage in Costa Rica, Malaysia); community projects and employment; carbon sequestration payments • Effectiveness: protected areas work where enforced but face illegal logging, mining, poaching, roads, conflict; drivers (population growth, demand for beef, soya, palm oil, timber, minerals) originate outside the forest; climate change cannot be controlled locally • Judgement: combination of large protected areas, economic alternatives for local people and reduced international demand needed; overall loss continues	9
10.2	• Carbon sequestration reducing climate change; genetic resources / medicines; rainfall regulation; global biodiversity	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 56-57; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
11.1	Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Physical features: mass / gravity retaining an atmosphere and giving pressure for liquid water; distance from the Sun and the Goldilocks zone; speed of rotation reducing temperature extremes; tilt producing seasons; magnetic field deflecting the solar wind; the Moon stabilising the orbit and producing tides • Importance of liquid water: solvent, transport, temperature control, anomalous expansion, high specific heat capacity, habitats, absorption of UV before the ozone layer • Development of life: Archaea at vents; photosynthesis; oxygen reacting with iron then accumulating; ozone layer allowing colonisation of land; aerobic organisms • Life changing conditions: carbon sequestration offsetting solar brightening, biogeochemical cycles, transpiration and rainfall, regulation of atmospheric gases • Methods of studying past conditions: proxy data, isotope ratios, ice cores, Mg:Ca ratios; understanding of continental drift and ocean currents • Links to current threats (climate change, ozone depletion) and a justified conclusion about the interdependence of physical and biological factors Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-9; AQA spec 3.1.1.	25

Question	Marking guidance	Marks
11.2	Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Ecological features: cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web; specific abiotic requirements (temperature, light, salinity, low turbidity) • Importance: quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism • Vulnerability: narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching; slow growth so slow recovery; physical damage; overfishing and food-web effects (crown of thorns); introduced species; dependence on mangroves and seagrass • Conservation: MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES • Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant • Evaluation of local vs global threats and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	25